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Foundations of Programming

Assignment No: 8

PROBLEM STATEMENT:

Write a program in **C** to accept marks of five courses of a student and compute the result. A student is considered **PASS** if he/she scores **40 marks or more in each course**. If the student passes, calculate the aggregate percentage and assign the grade as follows:

- Aggregate **≥ 75%** : **Distinction**
- Aggregate **≥ 60% and < 75%** : **First Division**
- Aggregate **≥ 50% and < 60%** : **Second Division**
- Aggregate **≥ 40% and < 50%** : **Third Division**

OBJECTIVE:

- To understand the use of **conditional statements** in C
- To apply **logical operators** for decision making
- To calculate **total, percentage, and grade** of a student
- To enhance problem-solving skills using real-life scenarios

THEORY:

In academic evaluation systems, student performance is assessed based on individual subject marks as well as overall aggregate performance. In C programming, such evaluation systems can be implemented using conditional statements like `if`, `else if`, and `else`. Logical operators help in checking multiple conditions simultaneously. Percentage calculation involves arithmetic operations, and grading logic is implemented based on predefined ranges. This program demonstrates structured programming and decision-making in C.

ALGORITHM:

PLATFORM: Linux

PROGRAM:

```
#include <stdio.h>


int main() {
    int m1, m2, m3, m4, m5;
    int total;
    float percentage;


    printf("Enter marks of 5 subjects:\n");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);


    if (m1 >= 40 && m2 >= 40 && m3 >= 40 && m4 >= 40 && m5 >= 40) {
        printf("Result: PASS\n");


        total = m1 + m2 + m3 + m4 + m5;
        percentage = total / 5.0;


        printf("Percentage = %.2f%%\n", percentage);


        if (percentage >= 75) {
            printf("Grade: Distinction\n");
        }
        else if (percentage >= 60) {
            printf("Grade: First Division\n");
```

```
        }
        else if (percentage >= 50) {
            printf("Grade: Second Division\n");
        }
        else {
            printf("Grade: Third Division\n");
        }
    }
    else {
        printf("Result: FAIL\n");
    }

    return 0;
}
```

```c
#include <stdio.h>

int main() {
    int m1, m2, m3, m4, m5;
    int total;
    float percentage;

    printf("Enter marks of 5 subjects:\n");
    scanf("%d %d %d %d %d", &m1, &m2, &m3, &m4, &m5);

    if (m1 >= 40 && m2 >= 40 && m3 >= 40 && m4 >= 40 && m5 >= 40) {
        printf("Result: PASS\n");

        total = m1 + m2 + m3 + m4 + m5;
        percentage = total / 5.0;

        printf("Percentage = %.2f%%\n", percentage);

        if (percentage >= 75) {
            printf("Grade: Distinction\n");
        }
        else if (percentage >= 60) {
            printf("Grade: First Division\n");
        }
        else if (percentage >= 50) {
            printf("Grade: Second Division\n");
        }
        else {
            printf("Grade: Third Division\n");
        }
    }
    else {
        printf("Result: FAIL\n");
    }

    return 0;
}
```

INPUT:

```
Enter marks of 5 subjects:
99 79 89 92 95
Result: PASS
Percentage = 90.80%
Grade: Distinction
```

80 75 70 85 90

OUTPUT:

Enter marks of 5 subjects:

80 75 70 85 90

Result: PASS

Percentage = 80.00%

Grade: Distinction

CONCLUSION: Thus, the program to calculate student result and grade based on marks was successfully implemented using C programming. This program demonstrates the effective use of conditional statements, logical operators, and arithmetic operations.

FAQs:

1. What is the objective of this C program?
2. Why is an if–else ladder used in this program?
3. Which data types are generally used in this program?